

HOMEWORK 6

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

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Part 1. Multiple Choice or True/False

1. (1 point) If a non-key column determines a key column, which normal form is violated?
A) 1NF
B) 2NF
C) 3NF
D) BCNF
2. (1 point) The essential condition of third normal form is automatically satisfied when
A) a table has only one non-key column
B) a table has only one key column
C) a table has only one primary key
D) a table has no foreign key
3. (1 point) (check all that apply) What is the relationship between the 3rd and Boyce-Codd normal form?
A) if BCNF is satisfied, then is the 3NF
B) If BCNF is violated, then 3NF is also violated
C) Both normal forms are identical
D) If 3NF is violated, then BCNF is also violated.
4. (1 point) (select all that apply) Which of following tables have the violation of the Boyce-Codd normal Form with the following assumptions:
 - 1) Each student may several advisors
 - 2) Each zip is for one city
 - 3) SSN is unique for all students
A) Students (SID, lname, fname, address)
B) Students (SID, lname, fname, Advisor, RelationshipRating)
C) students (SID, lname, fname, state, major)
D) students (SID, lname, fname, state, city, zip)
E) Students (SID, lname, fname, state, SSN)
5. (1 point) A transitive dependency is equivalent to which of the following?
A) A functional dependency between two or more key attributes.
B) A functional dependency between two or more nonkey attributes.
C) A relation that is in first normal form.
D) A relation that is in second normal form.
6. (1 point) In the ER model for a library checkout system, the involved entities include Book, BookCopy, Patron, Checkout, Return, Reservation, Employee, and Transaction. Assume each return is for many books. If a book is returned with damage, where do you record the information so that we can track the damage by user in case we need to?

- A) Book
 - B) BookCopy
 - C) Patron
 - D) Association between Return and BookCopy
 - E) Return
7. (1 point) The violation of the second normal form is often due to combining tables whose records are in
- A) 1:1 relationships
 - B) 1:m relationships
 - C) m:m relationships
 - D) O:O relationships
8. (1 point) The violation of the third normal form is often due to combining tables whose records are in
- A) 1:1 relationships
 - B) 1:m relationships
 - C) m:m relationships
 - D) O:O relationships
9. (1 point) (select all that apply) Boyce-Codd normal form would be violated if we combine tables whose records are in which of the following relationships?
- A) 1:1
 - B) 1:m
 - C) m:m
 - D) None of the above
10. (1 point) (check all that apply) What are the basic elements in a typical SQL DML statement?
- A) type of action
 - B) table names
 - C) column names
 - D) primary key
 - E) record filtering conditions